



HEXAGON

**MSC Nastran
Fluid-Structure Analysis,
Coupled Modes**

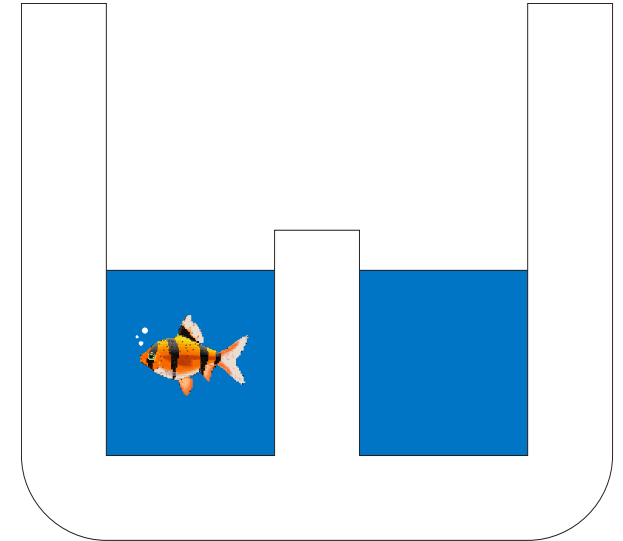
Overview

Fluid-Structure Analysis in MSC Nastran

- MSC Nastran has been able to solve fluid-structure problems for many years
- Fluid-structure analysis is a somewhat vague term
 - Vibro-acoustic analysis is sometimes called fluid-structure analysis
 - Fluid-structure analysis may involve fluid dynamics (CFD) or explicit analysis like sloshing or bird strike
 - The fluid may not even be modelled explicitly at all (e.g. boundary element methods)
- In MSC Nastran, 3 types of fluid-structure analysis are possible
 - Co-simulation with a separate CFD solver solving Navier-Stokes or Lattice Boltzmann equations
 - This requires MSC Nastran, an external CFD solver (e.g. SCFlow) and CoSim to federate the 2 solvers
 - SOL 700 to execute an explicit solution with the Dytran Eulerian solver, e.g. bird-strike
 - **Integrated coupled or weakly coupled, interior or exterior acoustics problems**
- In this presentation, we will only look at the second of these

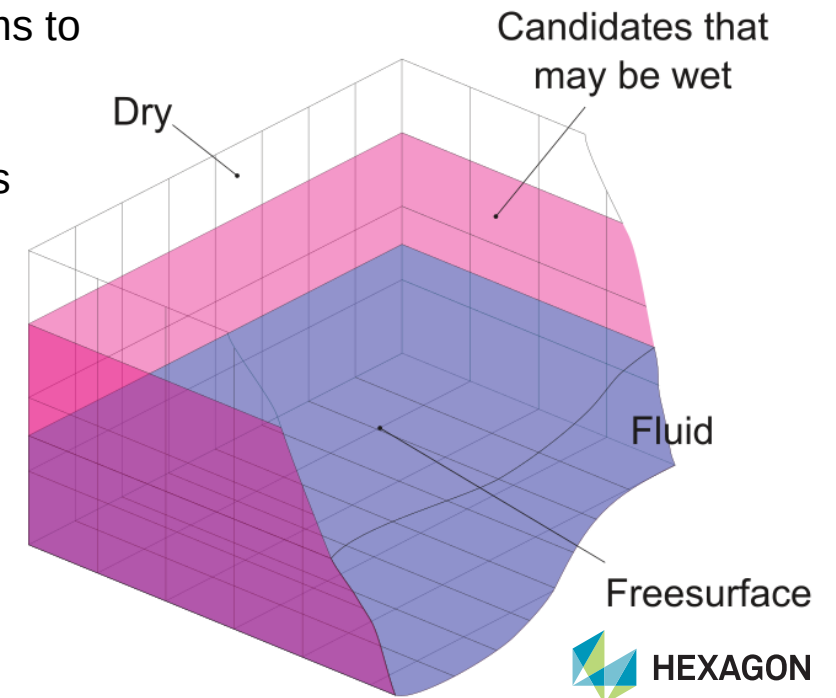
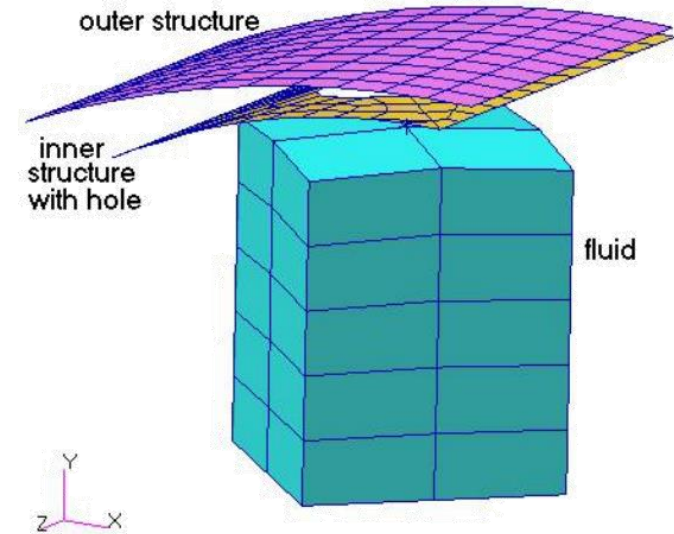
Fluid-Structure Analysis in MSC Nastran

- Fluid-structure analysis solves a six degree of freedom (DOF) structural problem defined with displacement variables coupled with a single DOF fluid cavity defined with pressure variables
 - Sometimes called coupled F-S or acoustic analysis
- In MSC Nastran there is an assumption of small displacements with interior or exterior fluids or both
 - Interior fluids are partially or wholly enclosed by a bounding structural mesh
 - Any fluid not touching the structure will be assigned a reverberant condition
 - Exterior fluids have their exterior faces defined with semi-infinite elements that define a non-reflecting boundary (sound radiation to the far field)
 - The fluids do not flow or exhibit large motion; sound pressure propagates through the fluid medium through pressure perturbations
 - The fluids may be a gas (like air) or a liquid (like water or gasoline)
 - Gas cavities are “light” fluids, while liquids get the moniker “heavy” fluids
- Fluid mesh or no fluid mesh?
 - For the boundary element method, the fluid interior is not meshed but is defined by where it touches the structure
 - For the other methods, the fluid media are meshed with solid finite elements and assigned MAT10 properties
 - The same interface is used for light or heavy fluids



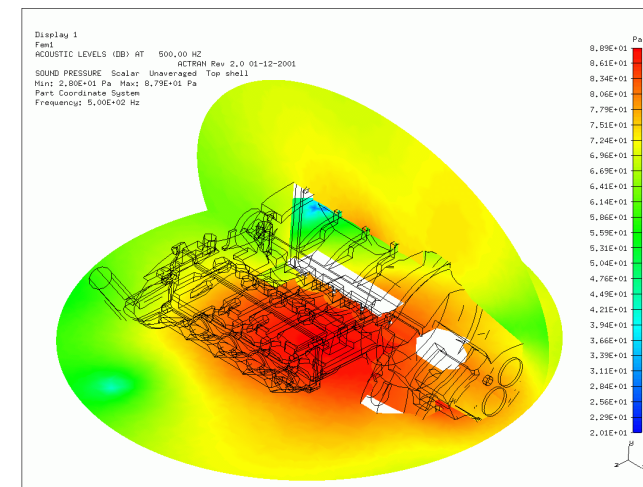
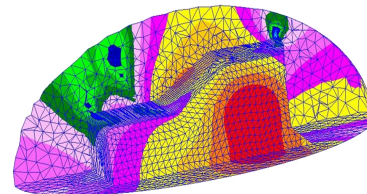
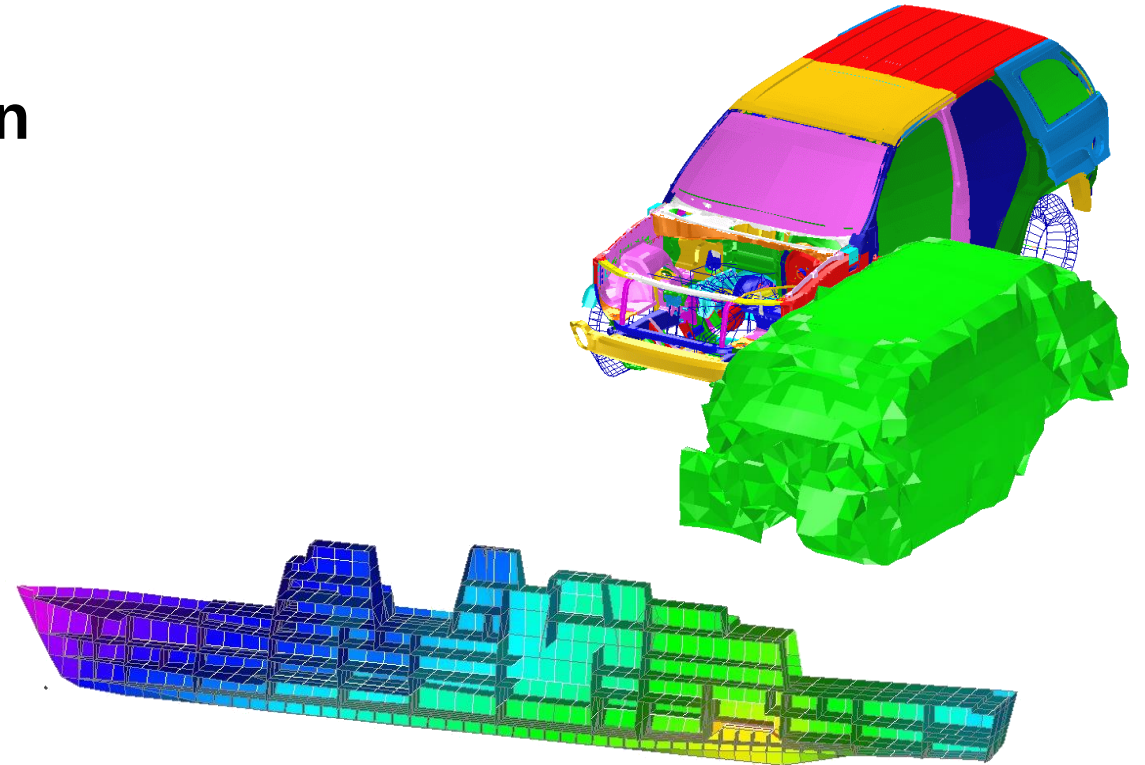
Fluid-Structure Analysis in MSC Nastran

- When the fluid is meshed with solid elements, a coupling surface is required to describe where the fluid touches the structure (the wetted surface)
 - This surface couples the 6 DOF structure displacements to the 1 DOF fluid pressures where the vibrating structure defines a velocity boundary condition on the fluid and the acoustic pressure in the fluid defines a pressure load on the structure
 - The coupling surface may be computed automatically by MSC Nastran, or provided by the user from some external source
 - The coupling may be checked visually using SETs of structure and fluid elements output by MSC Nastran – these SETs may be used to include/exclude items to moderate the coupling
- When the boundary element method is used, the potentially wetted surface is defined by the user, but the final wetted elements are governed by geometric parameters defining the fluid level and the orientation of the free surface
 - This capability is known as the virtual mass method and only accounts for the added mass the fluid has on the structure
- Both of these methods may exist in the same model, and both methods allow multiple independent fluid domains to be defined, each with its own characteristics (density, fluid level, etc.)



Fluid-Structure Analysis in MSC Nastran

- Applications of these method may include
 - Interior acoustics
 - Passenger, engine & boot (trunk) compartments of a car
 - Cabin of an aircraft
 - Engine, cargo & berth compartments of a ship
 - Anechoic testing chamber
 - Compressor housing
 - Exterior acoustics
 - Noise in the far field (at some remote distance from the noise source)
 - Semi-infinite elements provide the non-reflecting boundary condition on a convex surface



Solution Methods

Solution Methods

- Two solution methods are available to solve the linear coupled fluid-structure problem
 - Direct – the solution variables are the GRID point displacements and pressures and we speak of physical variables
 - Modal – a modal reduction phase is used to compute generalized DOFs to be used to solve the modally reduced problem, and we speak of modal variables
 - Once the modal problem is solved, the same modal basis is used to recover physical response quantities
 - Although modal reduction takes time, it is generally speaking faster than solving the direct problem, but it introduces an approximation associated with modal truncation
- Coupled fluid-structure analysis is not available for a pure statics problem (SOL 101), but a static problem may be solved to compute the normal modes problem under a preload, for example
- Fluid cavities may be present in a model with superelements with the restriction that the cavities may not be interior to internal superelements
 - A special superelement (PARAM,FLUIDSE) is reserved where all fluid cavities may be placed and they will be reduced
- External superelements may contain one or more fluid cavities

Solution Methods

- Direct methods where a fluid-structure model may be analyzed
 - SOL 103 – real normal modes
 - SOL 107 – direct complex eigenvalues
 - SOL 108 – direct frequency response
 - SOL 109 – direct linear transient response
 - SOL 200 – multi-disciplinary analysis, sensitivity and optimization
 - SOL 400 – multi-disciplinary nonlinear analysis with linear perturbations*

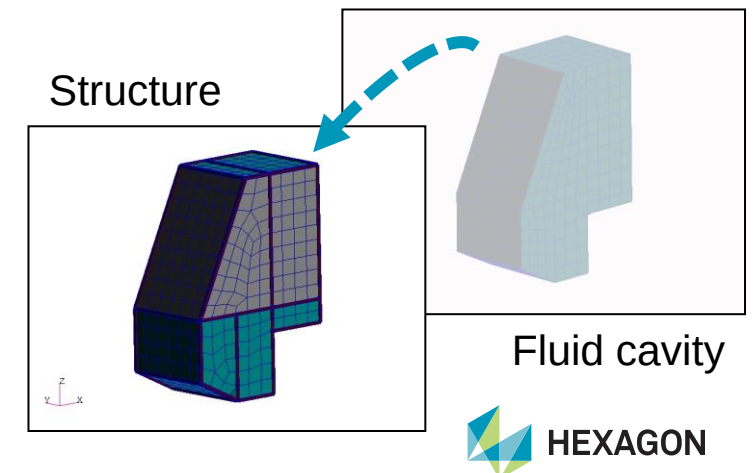
Physical Space
- Modal methods where a fluid-structure model may be analyzed
 - SOL 110 – modal complex eigenvalues
 - SOL 111 – modal frequency response
 - SOL 112 – modal linear transient response

Modal Space
- * The nonlinear problem is solved directly and a linear perturbation about a nonlinear deformed shape may be carried out using either direct or modal methods

Uncoupled and Coupled Modes

Uncoupled Modes

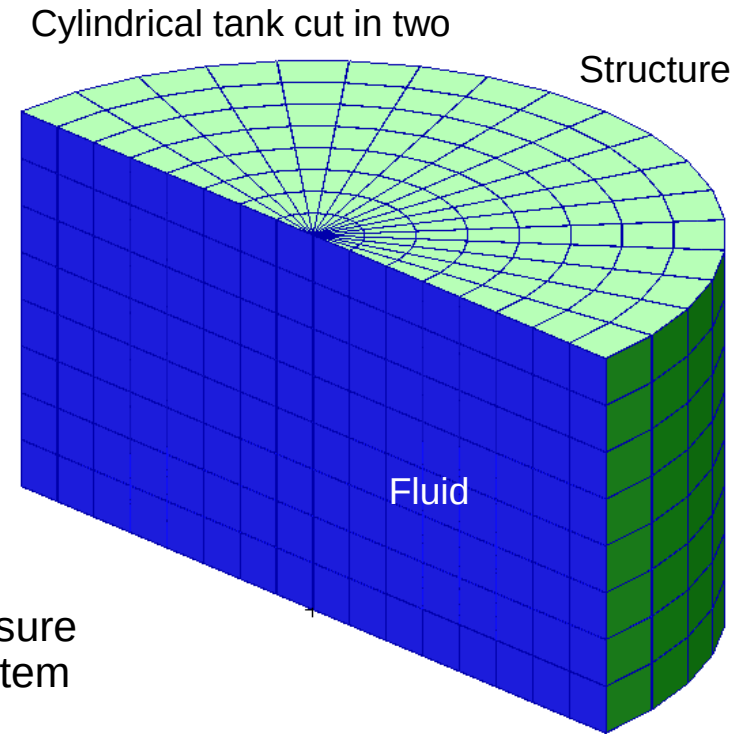
- Fluid-structure analysis with light fluids (gases) - uncoupled modes
 - The fluid mesh of solid elements is assigned MAT10 properties to define its bulk modulus, density and the speed of sound in the gas
 - The modes of the structure are computed as if the fluid were not present (in vacuo)
 - The modes of the fluid are computed as if the structure were not present (rigid free faces = perfect reflection)
 - The presence of the gas does not change the behaviour of the structure and the presence of the structure does not change the behaviour of the fluid(s) in any significant way
 - The modes may be computed using a real symmetric eigenvalue solver as in SOL 103
 - For forced response analysis using uncoupled modes, the physical coupling between fluid and structure is modally reduced using the real modes of the fluid and structure
 - It is possible to select a hybrid direct-modal solution where either only the structure, or only the fluid, is modally reduced
 - As the modes of the structure and fluid are computed separately, we refer to these separate modal bases as uncoupled modes



Coupled Modes

- Fluid-structure analysis with heavy fluids (liquids) – coupled modes
 - Heavy fluids like water or gasoline change the dynamic behaviour of the structure that contains them and the behaviour of the fluid is affected by the surrounding structure
 - A heavy fluid contained by a structure has a stiffening effect on it
 - The inertia of the heavy fluid increases the apparent mass of the structure
 - The modes of the fluid-structure problem are coupled and the system modes must be computed together – these are the coupled modes
 - For a meshed fluid, the coupling surface A between the fluid described with pressure variables and the structure described with displacement variables makes the system matrices asymmetric
 - With structural stiffness K , fluid mobility (inverse mass) H , structural mass M and fluid compressibility (inverse stiffness) Q , the coupled modes problem consists in solving the asymmetric eigenvalue problem – this has to be done efficiently
- Asymmetric matrices means the modes cannot be computed directly using a real symmetric eigenvalue solver
- It is possible to compute coupled modes using a complex eigenvalue solver, like in SOL 107 or 110
 - Although the problem is asymmetric, in the absence of damping, both the eigenvalues and eigenvectors of the coupled modes are real valued
 - However, SOL 107/110 both output complex quantities with zero imaginary parts, and solution times can be long
 - There is no way to use the complex modal basis for a reduced dynamic response solution

$$\left(\begin{bmatrix} K & -A \\ 0 & H \end{bmatrix} - \lambda \begin{bmatrix} M & 0 \\ A^T & Q \end{bmatrix} \right) \begin{Bmatrix} u \\ p \end{Bmatrix} = 0$$



Why Do We Need Coupled Modes?

- Among the many assumptions for structure only analysis is that we ignore the presence of the air (fluid) around the structure
 - For stiff structures like cars, aeroplanes, bridges this assumption is tenable
 - We tend to add the air only when we want to know the pressure levels
 - For light (flexible or flimsy) structures like thin film parabolic reflectors intended for use on satellites in space, testing here on earth must account for the presence of the surrounding air as it alters the structure's behaviour
 - For liquids (heavy fluids), the presence of the fluid cannot usually be ignored
- When heavy fluids touch the structure, account for additional effects is generally needed
 - If the structure is very stiff, it may only be necessary to account for the added mass of the fluid
 - MFLUID (also called the virtual mass method) may be used – this is a boundary element method
 - If the structure is stiff, but not too stiff, additional effects may be important
 - The structure may experience added mass from the presence of the fluid (as with MFLUID) – this will tend to decrease the mode frequencies when compared with an empty vessel
 - The fluid may experience added compressibility (fluid mass) due to the presence of the flexible structure – this will tend to decrease the mode frequencies when compared with those computed with the fluid alone (where a rigid wall is assumed)
 - Any constant pressure modes in the fluid will increase the stiffness of the structure due to the fluid resisting a change in its volume – any modes dominated by structure behaviour will increase their frequency
 - Any rigid body modes in the structure will increase the mobility of the fluid (added fluid “stiffness”) – any modes dominated by the fluid behaviour will increase their frequency

Coupled Modes in MSC Nastran

- In MSC Nastran V2019FP1, a real eigenvalue capability was added for solving the coupled modes problem
 - Available in SOL 103 (real modes), SOL 111 (modal frequency response) and SOL 112 (modal transient response) for the computation of coupled modes of the fluid-structure problem
 - No modelling changes are required to use coupled modes
 - Residual vectors may be computed for modal forced response analysis (SOL 111/112) using the coupled modal basis
 - The direct methods of forced response remain available especially useful for comparisons
- In MSC Nastran V2021, component mode synthesis using coupled modes for an external superelement containing one or more heavy fluid cavities was added
 - Coupled modes are used to reduce the external superelement, containing a heavy fluid cavity, to a Craig-Bampton boundary representation; i.e. the modal reduction is carried out using coupled modes
 - No modelling changes are required to use coupled modes
 - Boundary matrices and OTMs are used in the usual way for the assembly run
 - No fluid points are allowed in the boundary and sufficient DOF must be placed in the B set (fixed boundary CMS) of the external superelement to remove any rigid body motion
 - B set CMS is recommended. If you need boundary DOF to remain free during the computation of the component modes, place these DOF on Access Points (ACCSPT)
 - If damping is to be defined for the external superelement, only modal damping or structural damping may be used

Coupled Modes in MSC Nastran

- In MSC Nastran V2021.1
 - Coupled modes for EXB external flexible body generation was made available for the AVL Excite software – this complements coupled modes for Adams MNF flexible body generation that was made available in the previous release
 - The exterior acoustics capability was extended to add ACOWEAK,YES to the list of possible analysis capabilities supported by coupled modes
 - An accelerated method was added for the coupled modes capability leveraging ACMS to compute real uncoupled modes as starting vectors for the eigenvalue computation
- In MSC Nastran V2023.4
 - A new method for coupled modes of very large-scale problems was added
 - Both the original and new methods are available for efficient processing adapted to the size of the problem to be solved
 - The user decides which method they wish to use via case control commands

Setup and Inputs

Setup and Inputs

- For SOL 103, 111, 112 and 200
 - For small to medium sized models, use METHOD(COUPLED)=
- ACMS may be used with this method, and then a METHOD(STRUCTURE) and METHOD(FLUID) command are also required
- For very large models, use METHOD(SYMCOUPL)=
- ACMS is not available with this method

```
SOL 103
METHOD (COUPLED) =10
BEGIN BULK
EIGRL, 10, , , 20
...
ENDDATA
```

```
DOMAINSOLVER ACMS
SOL 103
METHOD (STRUCTURE) =51
METHOD (FLUID) =52
METHOD (COUPLED) =10
BEGIN BULK
EIGRL, 10, , , 20
EIGRL, 51, , 400.
EIGRL, 52, , 200.

...
ENDDATA
```

```
SOL 103
METHOD (SYMCOUPL) =10
BEGIN BULK
EIGRL, 10, , , 20
...
ENDDATA
```

Setup and Inputs

- For SOL 103, 111, 112 and 200
 - If a static preload is required, reference it in the usual way with STATSUB(PRELOAD)

```
SOL 103
DISP (PLOT) =ALL
SUBCASE 10
  LOAD=4
SUBCASE 20
  STATSUB (PRELOAD) =10
  METHOD (COUPLED) =10
...
BEGIN BULK
EIGRL, 10, , , 20
FORCE, 4, 92, , 1., 0., 100., 0.
...
ENDDATA
```

Setup and Inputs

- For METHOD(COUPLED), two methods are available to solve the eigenvalue problem, either the right eigenvector method (MF method) or its adjoint left eigenvector problem (SM method)

$$\left(\begin{bmatrix} K & -A \\ 0 & H \end{bmatrix} - \lambda \begin{bmatrix} M & 0 \\ A^T & Q \end{bmatrix} \right) \begin{Bmatrix} u_R \\ p_R \end{Bmatrix} = 0 \qquad \begin{Bmatrix} u_L \\ p_L \end{Bmatrix}^T \left(\begin{bmatrix} K & -A \\ 0 & H \end{bmatrix} - \lambda \begin{bmatrix} M & 0 \\ A^T & Q \end{bmatrix} \right)^T = 0$$

- These 2 methods have different formulations of the symmetric eigenvalue problem and lead to more efficient solution depending on the characteristics of the model to be solved
 - The method is chosen automatically according to the number of rigid body and/or constant pressure modes and sizes of the fluid & structure domains
 - Identical eigenvalues are obtained with the 2 methods
 - The eigenvectors are normalized slightly differently in the 2 methods for efficiency
- For METHOD(SYMCOUPL), the method performs faster if no constant pressure modes are present

Setup and Inputs

- Forced response (SOL 111 or 112) with coupled modes only requires attention to be paid to modal damping (if defined)
 - With coupled modes, the modal damping is a function of the coupled modal basis, and there is now only a single set of eigenvalues and eigenvectors for the coupled problem (instead of 2 separate modal bases)
 - Requesting SDAMPING(STRUCTURE) or SDAMPING(FLUID) will result in no damping
 - Instead use SDAMPING(COUPLED)=

```
SDAMPING (COUPLED) =15
```

```
...
```

```
BEGIN BULK
```

```
...
```

```
TABDMP1,15
```

```
,0.0,0.02,100.0,0.0,0.04,ENDT
```

```
...
```

```
ENDDATA
```

Data Recovery

- Eigenvector output is available in the usual way using the DISPLACEMENT or VECTOR output request (they do the same thing)
 - The default is to output the eigenvectors for the structure and the fluid together in the same datablock mode by mode; for example

DISP=ALL

- will output both the structure and fluid eigenvectors in the same datablock
- If separate output of the structure or fluid or both eigenvectors is required, use one of the keywords STRUCTURE, FLUID or BOTH, for example

DISP (PRINT , STRUCTURE) =ALL will output only the structure eigenvectors

DISP (PRINT , FLUID) =ALL will output only the fluid eigenvectors

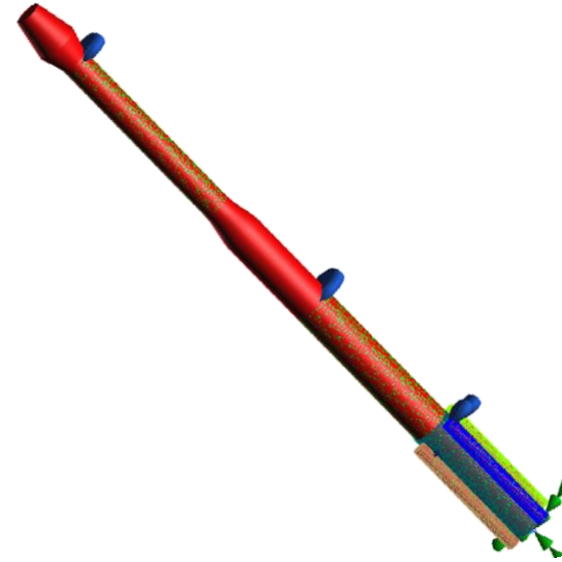
DISP (PRINT , BOTH) =ALL will output both the structure and fluid eigenvectors in separate datablocks

- For forced response, all the usual outputs are available

Typical Applications for Coupled Modes

Applications

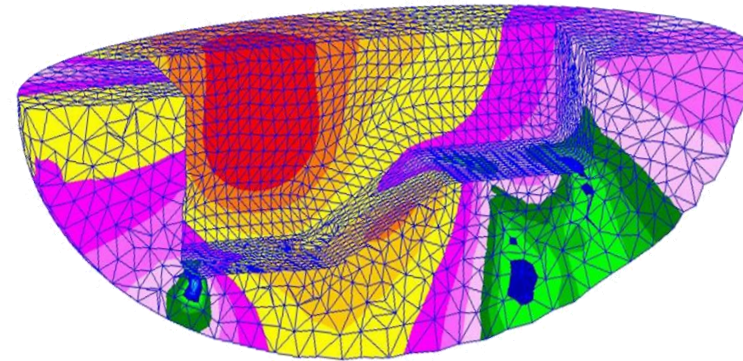
- Liquid filled reservoirs, for example
 - Hydrazine in rocket launchers and satellite boosters



- Gasoline tank in an automobile



- ICE oil filled sump
 - Correct vibration modes required for NVH
 - Exterior acoustic field for sound radiation to the far field (e.g. drive past noise)



End